NUS GRIP
Ambient Systems

Climate-as-a-Service for Modern Indoor Environments

Assignment #6
October 27, 2020

Introduction

Vision: create a safe, healthy, and productive indoor environment for everyone.

Mission: become a leading global provider of comprehensive smart building solutions.

Goal: develop new products enabled by IoT and that delight property owners and building operators, starting with the indoor farming segment.



Overview

1. Problem
2. Solution
3. Product/Technology/IP
4. Market
5. Business Model
6. Competitive Analysis
7. Financials
8. Milestones
9. Traction
10. Team

Section 1: Problem



Problem Statement



Climate Change

Need for greater control over indoor environments



Urban Pollution and Indoor Contaminants

Need for cleaner air



Rising Energy Costs

Need to keep energy consumption low



Increasing Technological Complexity

Need for simple solutions

Top Five Pain Points of Property Owners

Top **five pain points** of property owners and building operators across building types:

1. Hard to find trained personnel to continuously manage building's indoor environment
2. No or minimal visibility into real-time environmental parameters
3. Outdated and incompatible legacy systems
4. Competing priorities (e.g., improve environmental parameters or reduce energy costs)
5. Evolving legal compliance and certification requirements

Key takeaway: building operators face similar problems even when accounting for differences in building use.

Section 2: Solution

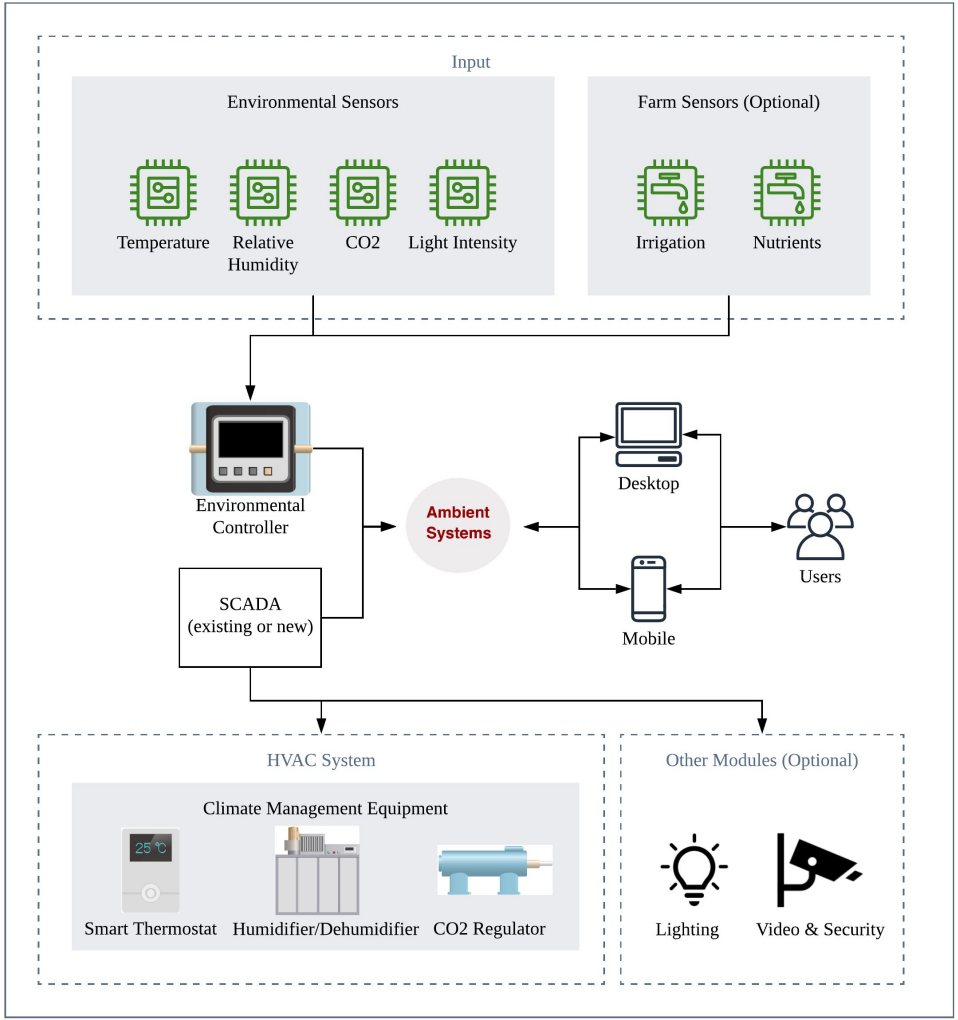


Three-part solution



IoT	IoT and ML Platform	Service Model
• Temperature sensor ○ (0 to 50 C +/- 0.1 C) • Relative humidity sensor ○ (0% to 100% +/- 1%) • CO2 sensor ○ (ppm +/- 5 ppm) • Light intensity sensor • Relevant connectors, mountings, gateway, cabling, and controls	• Real-time data delivered using modern web and mobile apps • Provides alerts via email or text when environmental conditions are beyond optimal range • Provides recommendations to remedy the environmental conditions	• Provide “Climate-as-a-Service” • User pays an annual subscription fee based on farm type and area • Streamline financing of the CAPEX and OPEX of the system

Ambient MVP High-level Schematic



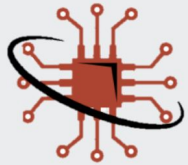
Section 3: Product & Technology



Ambient Systems

Building Automation for Indoor Farms

See how it works



Sign in

Email

brucewayne@mail.com

Password

SIGN IN

Forgot your password?

Ambient Sytems

Username



Farm #1

38 College Ave, Singapore

- Overview
- Yield Prediction
- Analytics
- ACMV Optimisation
- Maintenance
- Marketplace
- Settings

Overview

Camera



Yield Prediction

Crop Yield

Yield Week

13

1.7717 kg/m2

Crop Quality

Yield Week

13

1.7717 kg/m2

Anticipated Harvest Time

December 2020

Mo	Tu	We	Th	Fr	Sa	Su
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	1	2	3	4	5

Climatic Conditions

TEMPERATURE	RELATIVE HUMIDITY	CO2	LIGHT INTENSITY
20.5	55%	1050	600

Energy Consumption



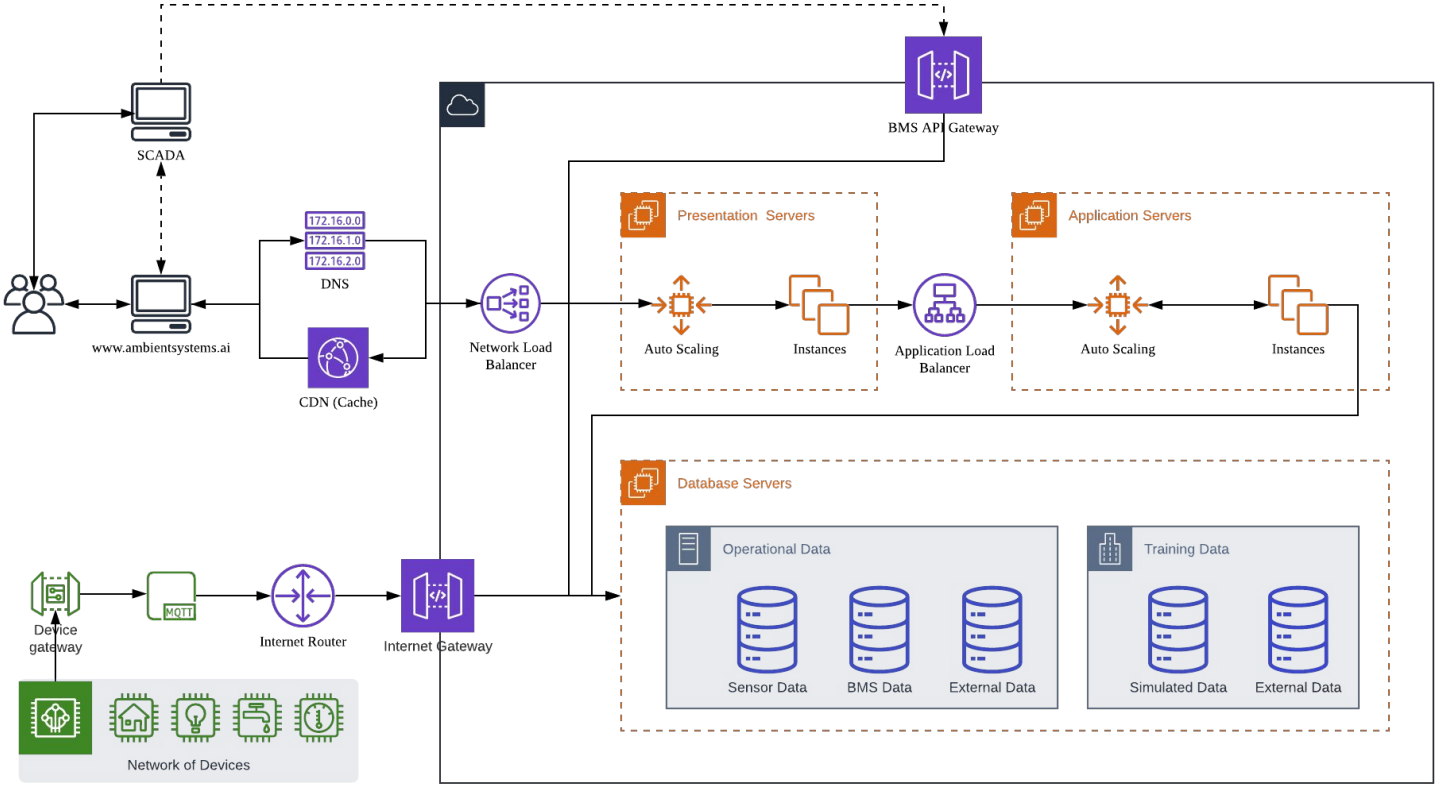
AC	16%
Humidifier	22%
CO2 Regulator	11%
Ventilation	39%
Lighting	12%

Bill of Materials: IoT Sensors

Component Name	Part Number	Manufacturer	Description	Picture	Unit Price	Supplier	Quantity	Total
Board and microcontroller	Arduino Uno Rev3 SMD -- A000066	Arduino	Board with processor and attachment ports		$29.86	RS-Online	2	$59.72
Temperature Sensor	MAX31889ALT+	Maxim Integrated	Board Mount Temperature Sensors 0.1 degrees C accuracy, 1-Wire Temperature Sensor		$4.40	Mouser	2	$8.80
Relative Humidity Sensor	SHT31-DIS-F2.5kS	Sensirion	Board Mount Humidity Sensors RH/ T Sensor		$9.11	Mouser	2	$18.22
CO2 Sensor	T6713-6H	Amphenol Advanced Sensors	Air Quality Sensors CO2 Sensor Module for AAS-AQS-UNO		$122.54	Mouser	2	$245.08
Light Sensor	TSL2561T ams	ams	TSL2561T ams, Ambient Light Sensor Unit Surface Mount 6-Pin TMB		$4.41	RS-Online	2	$8.82
Transceiver	LR1110IMLTRT	Semtech	RF Transceiver LONG RANGE WI-FI LOC LORA TRANSCEIVER		$14.93	Mouser	2	$29.86
ADCs	AD7795BRUZ	Analog Devices	Analog Devices AD7795BRUZ, 16-Bit Serial ADC Differential Input, 24-Pin TSSOP		$16.02	RS-Online	8	$128.16
Device enclosure	NSYTBP342912T	Schneider Electric	Plastic Enclosure, Flame Retardant, Multipurpose, Polycarbonate, 341 mm, 291 mm, 128 mm, IP66		$106.51	Element 14	2	$213.02
							Total	**$720 for 2 sensors**

Detailed bill of materials for the OEM MVP path has been been made for IoT Sensors

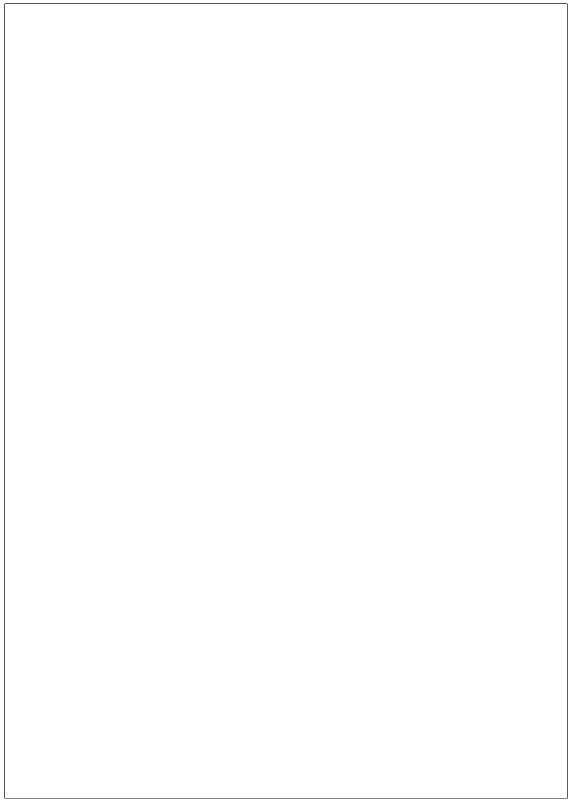
Preliminary System Architecture



MVP: Model

- System lays down the foundation for the prediction and optimization model
- Numerous options (depends on requirements)
- Potential: neural network (supervised learning)
- Consists of several “layers” of data -- potential structure:
  - Layer one: sensor measurements (i.e., raw sensor data)
  - Layer two: equipment operational logs
  - Layer three: output (yield, harvest time, crop quality)
- Use-cases:
  - Receive recommendations for climate controls (e.g., adjust temperature, increase humidity)
  - Predict crop output, harvest time, and quality
  - Automate climate controls

IP and



is an Intelligent Building Automation System, Method, and Apparatus for Smart Building Applications, which:

- Functions as an intelligent building management and automation system
- Manages indoor climate using Internet of Things (IoT) and
- Uses a diverse real-time dataset collected from an efficient and secure IoT network of sensors
- Trains -enabled models from IoT data to predict climate conditions and optimize the environment
- Provides a dashboard with real-time data to increase visibility into the climate conditions, along with user alerts when climate is not within target range
- Connects with existing Supervisory Control and Data Acquisition (SCADA) system used for building management
- Provides instructions to Heating, ventilation, and air conditioning (HVAC) equipment connected to SCADA
- Automates the building's HVAC system and related equipment to bring the indoor air quality within the desired goal

What makes Ambient unique?

Vertically integrated → collects the right data

Service → only one to offer simple yet comprehensive Climate-as-a-Service (CAAS)

→ database and computational process design

Section 3: Market



Total Available Market (TAM)

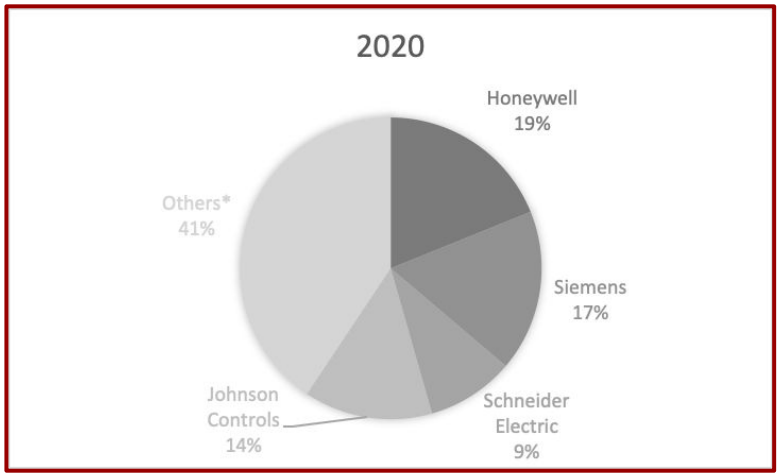
$4.74B

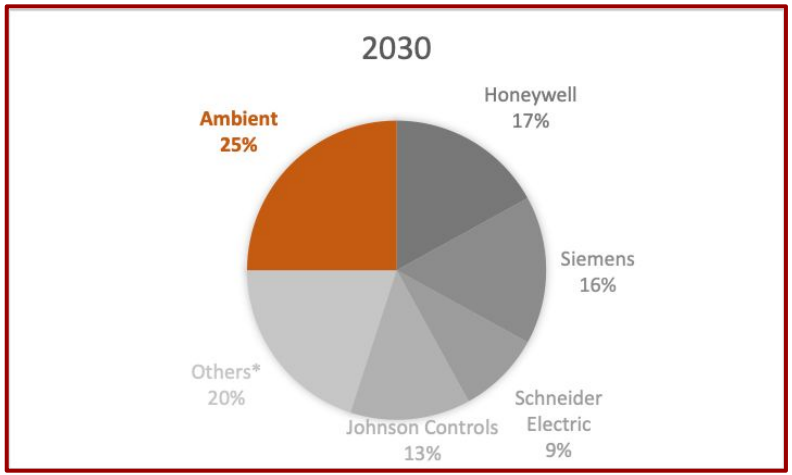
Total BAS Market: Revenue Forecast by Region, Global, 2017–2026

Year	Europe ($ Million)	North America ($ Million)	APAC ($ Million)	Latin America ($ Million)	MEA ($ Million)
2017	1,776.0	1,232.9	1,107.2	106.9	197.2
2018	1,823.5	1,269.9	1,149.2	111.0	203.5
2019	1,865.5	1,310.5	1,195.2	115.6	210.0
2020	1,878.5	1,321.3	1,210.2	117.7	212.9
2021	1,927.9	1,373.3	1,282.1	125.8	225.2
2022	1,963.6	1,420.8	1,340.7	131.5	234.1
2023	2,007.4	1,472.3	1,408.1	137.0	242.8
2024	2,059.8	1,523.4	1,483.4	143.2	253.3
2025	2,121.4	1,584.8	1,560.5	149.4	263.5
2026	2,184.8	1,655.2	1,646.5	155.4	274.9
CAGR	**2.3%**	**3.4%**	**4.7%**	**4.3%**	**3.9%**

Data from Frost & Sullivan

Serviceable Obtainable Market (SOM)

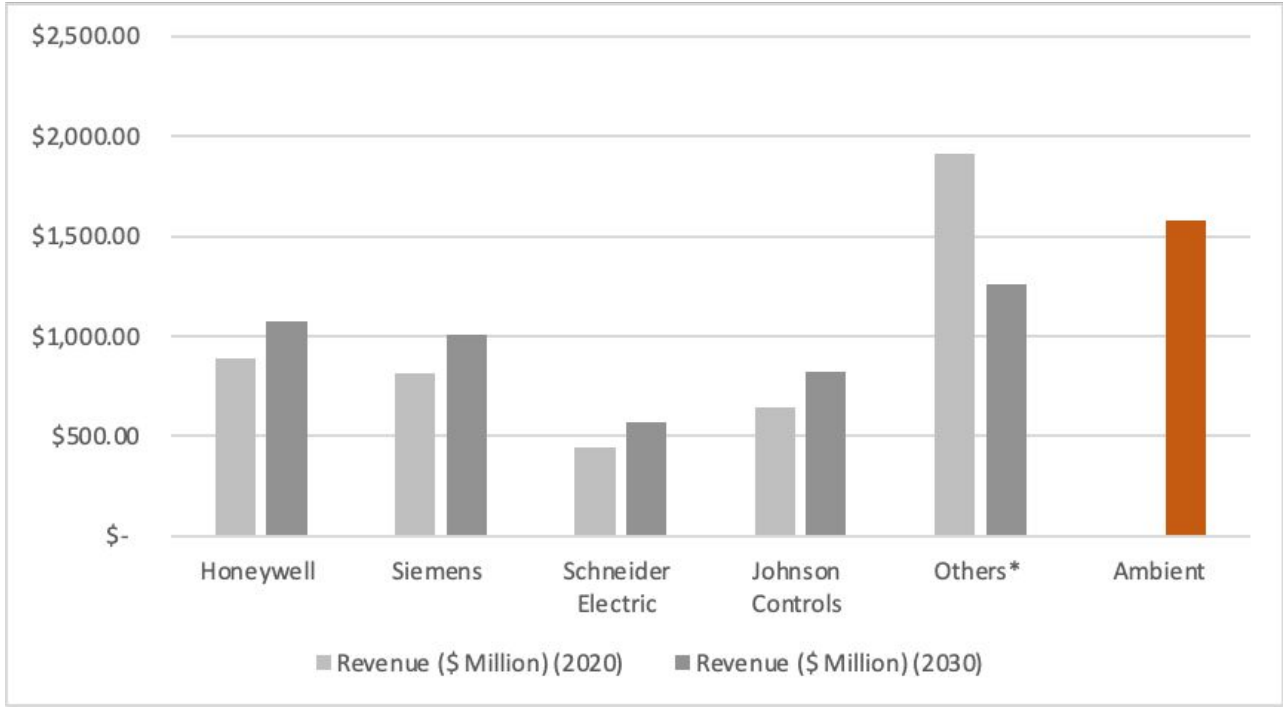




Data from Frost & Sullivan

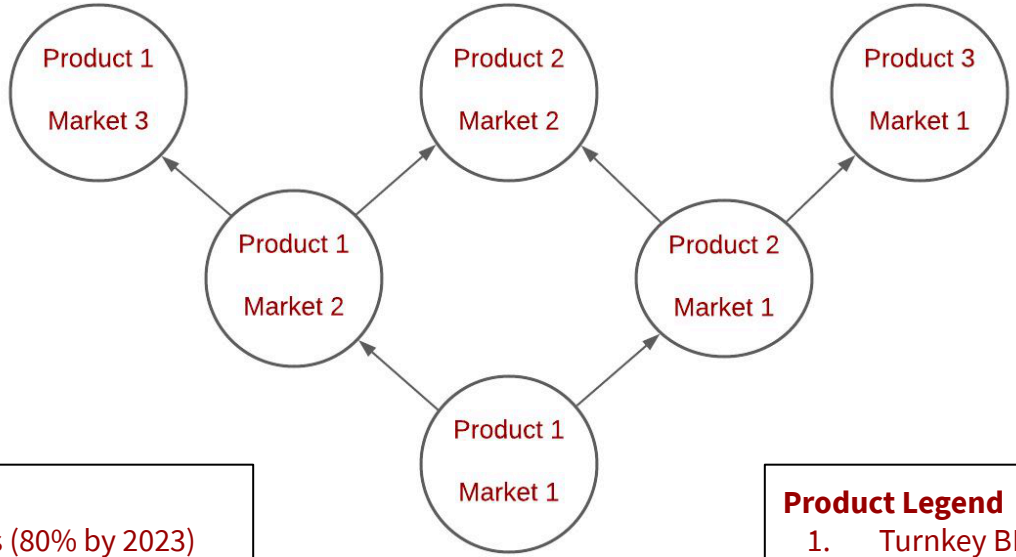
Serviceable Obtainable Market (SOM)

$1.57B



Data from Frost & Sullivan

Bowling Pin Diagram



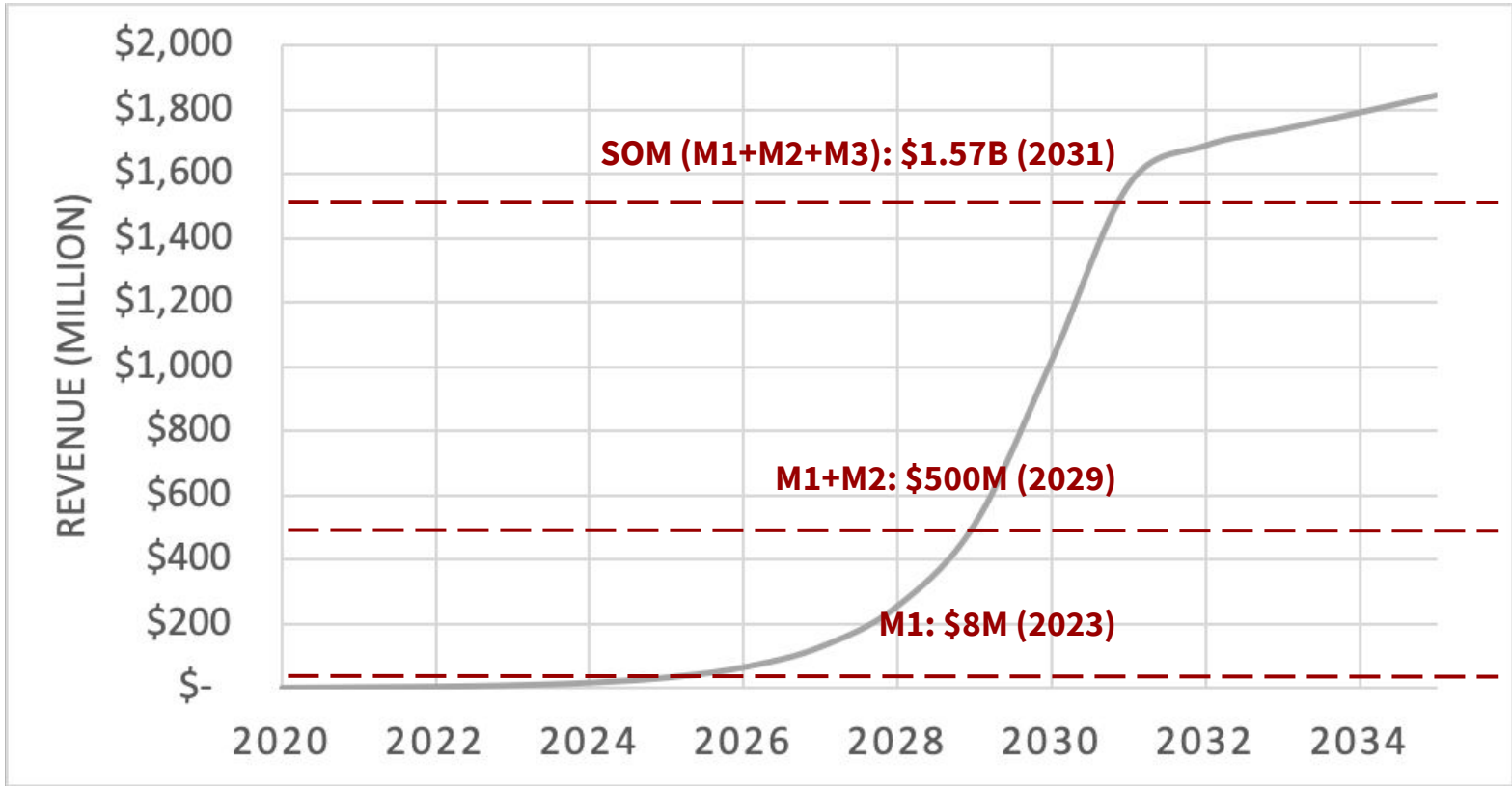
Market Legend

1. M1: SG Indoor Farms (80% by 2023)
2. M2: APAC C&I (50% by 2029)
3. M3: Europe C&I (50% by 2031)

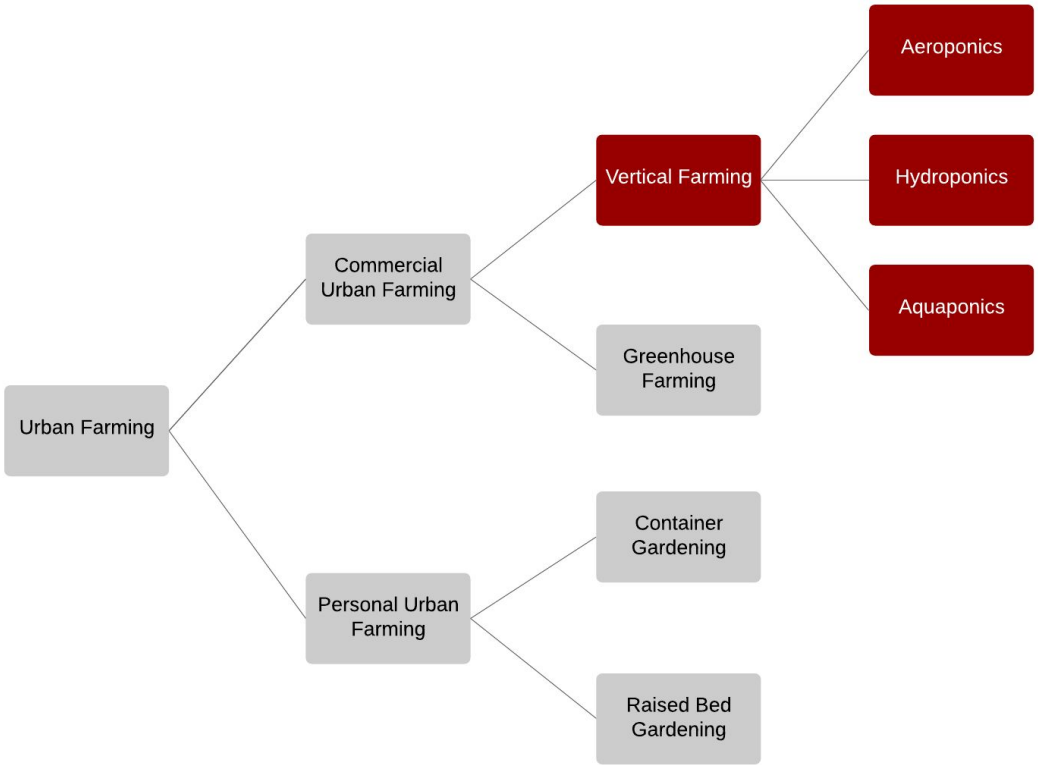
Product Legend

1. Turnkey BMS (system integrator)
2. enabled BAS
3. enabled

3s Framework

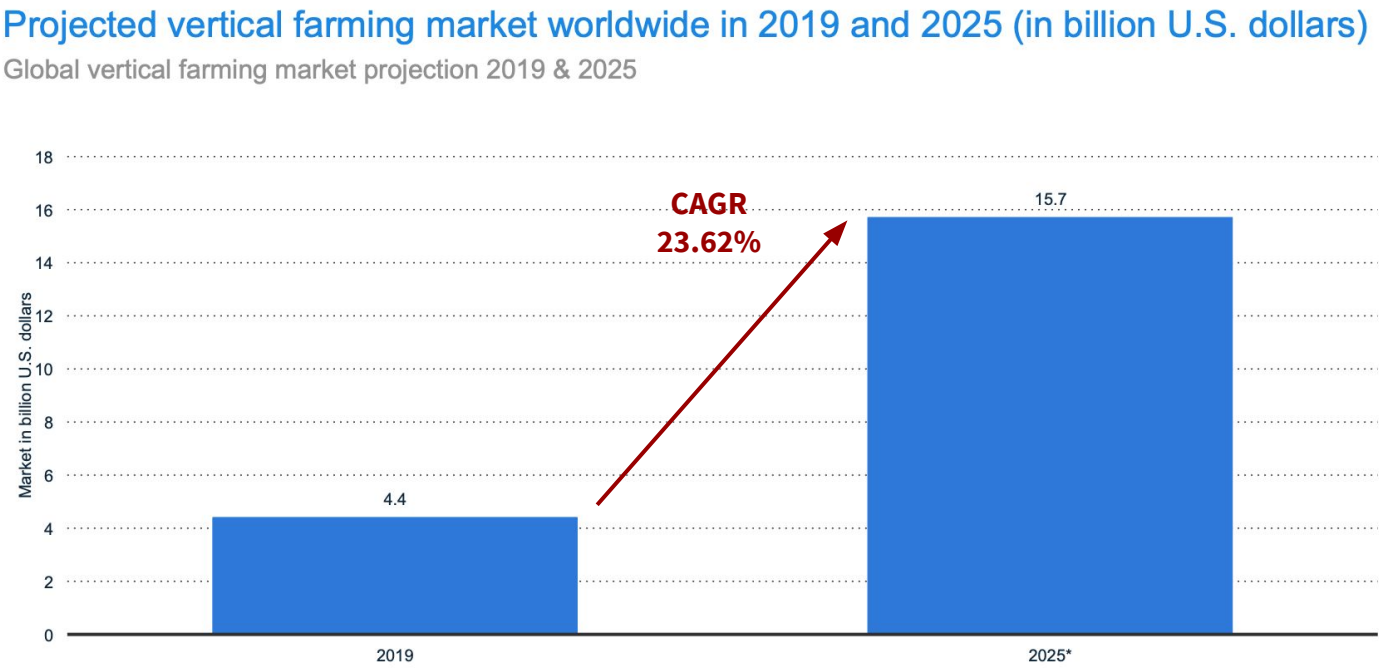


Urban Farming Market Segmentation



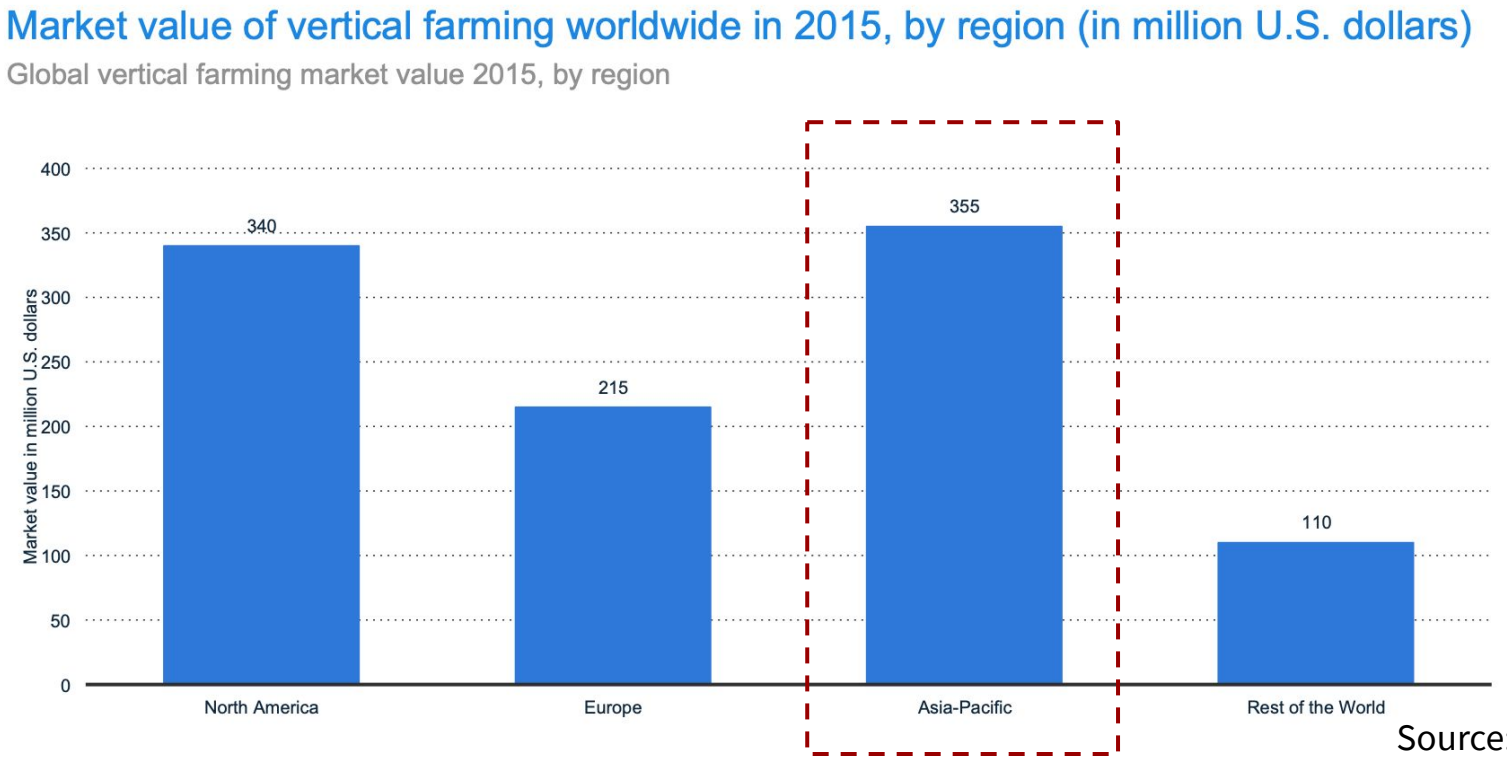
Source: Frost & Sullivan

Vertical Farming Market



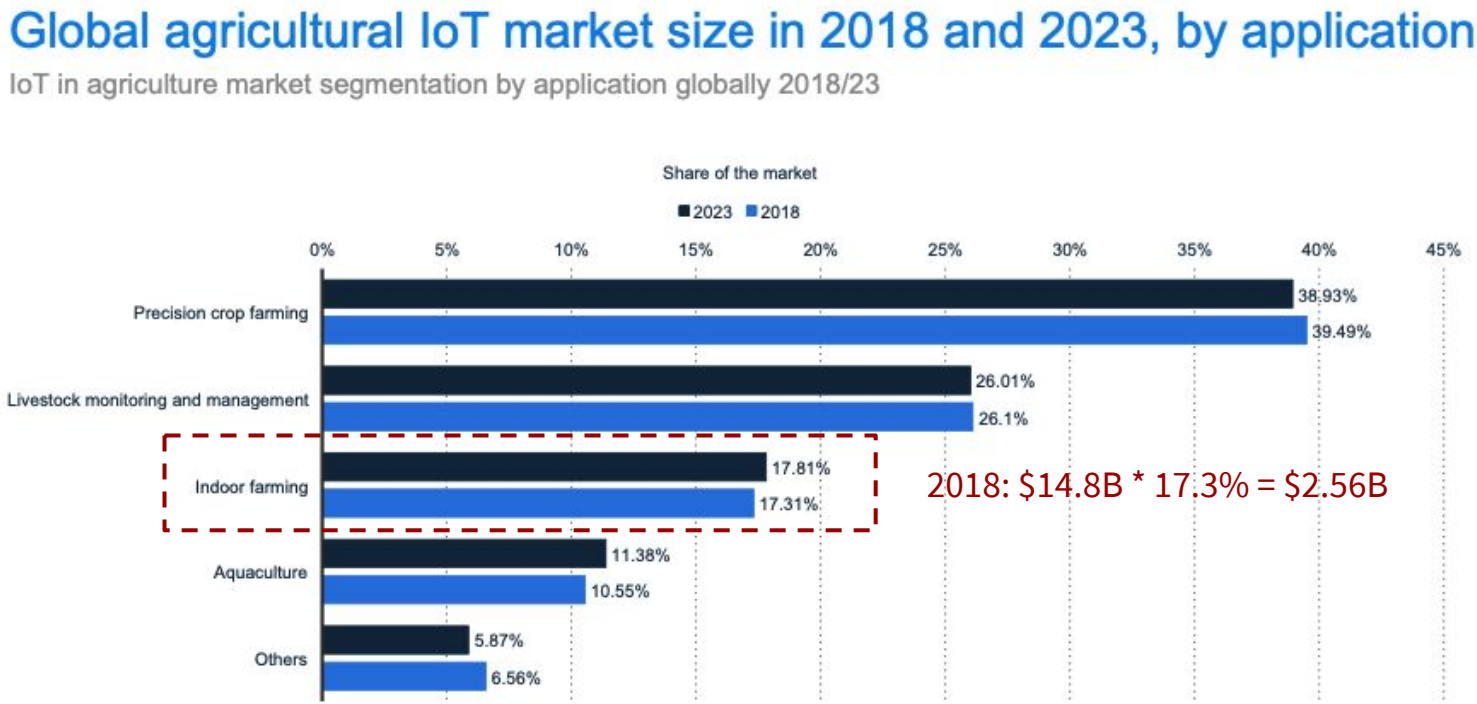
Source: Statista

Vertical Farming Market



Source: Statista

Agricultural IoT Market



2018: $14.8B * 17.3% = $2.56B

Source: Statista

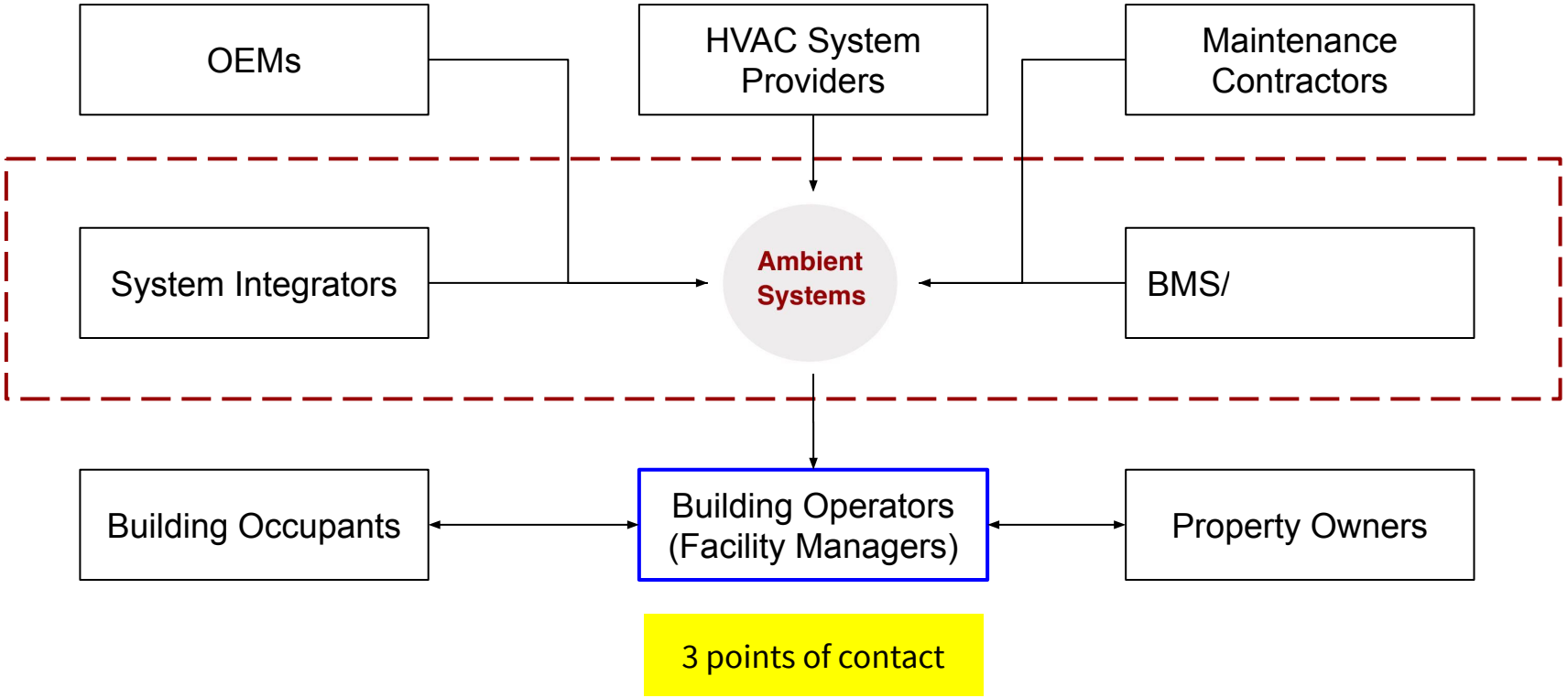
Vertical Farming is a new and rapidly growing industry.

The potential market that can be addressed is greater than $1B.

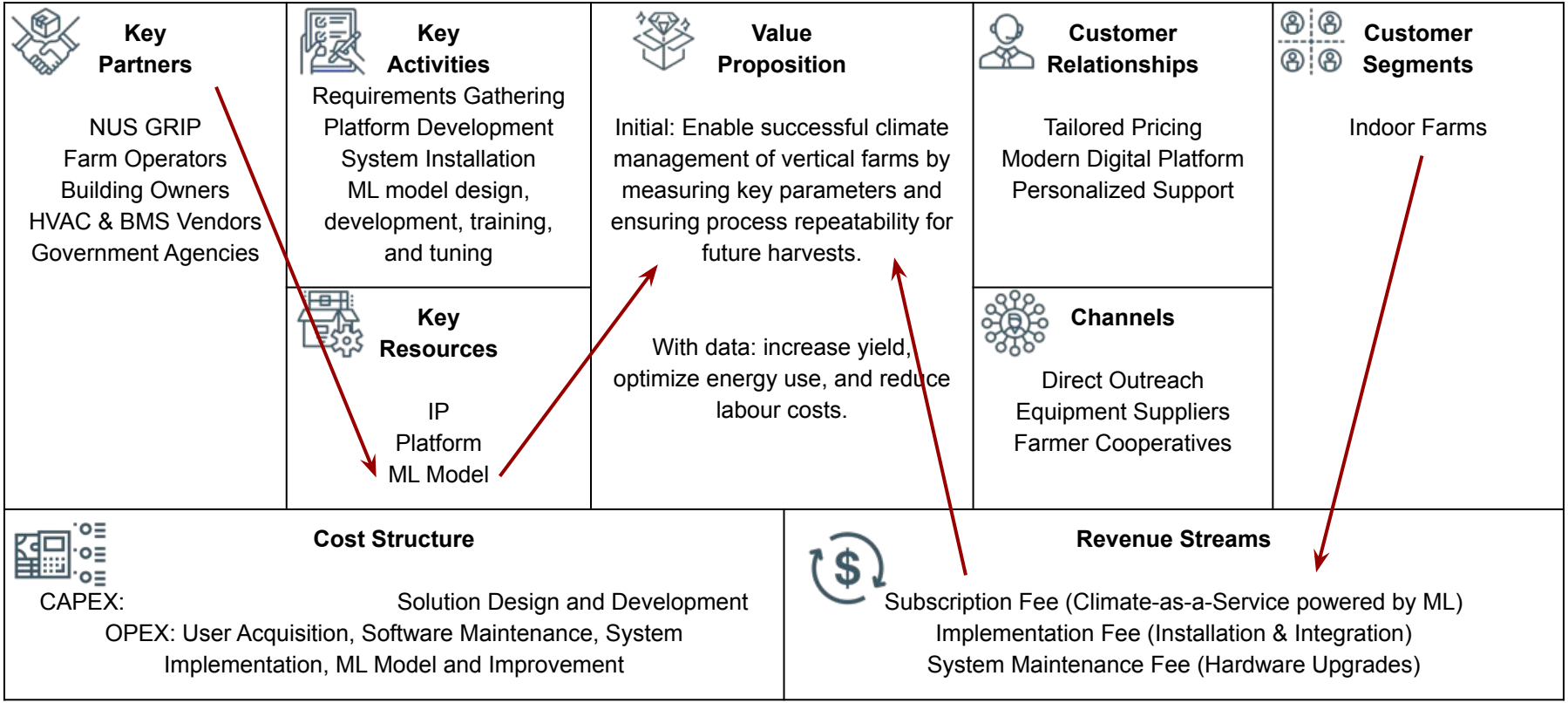
Farm operators have indicated interest and need for this type of solution.

Section 5: Business Model

Value Chain Analysis



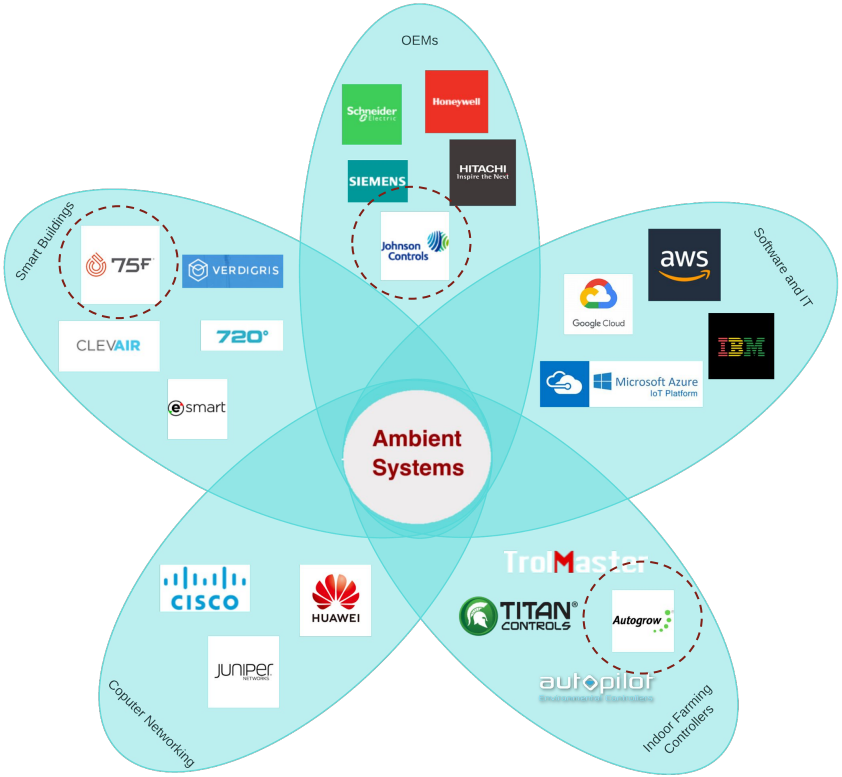
Business Model Canvas



Section 6: Competitive Analysis



Competitive Analysis: Petal Diagram



Ambient is positioned at the intersection of multiple segments in order to provide a modern building automation system (BAS) that is tailored for the indoor farming market.

Competitive Benchmark

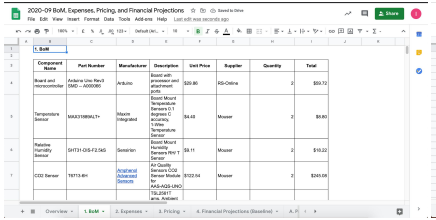
	Ambient	Autogrow	75F	Johnson Controls
Segment	Multiple	Indoor Farming Controller	Smart Buildings	OEM
Proximity to Customer	✓ Singapore	✗ US	✗ US	✓ International
Specific vs. Generic	✓ Indoor Farms	✓ Indoor Farms	✗ No	✗ No
Modern Web-based UI	✓ Yes	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ Yes	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ ML models	✗ No	✗ Some	✗ Some
Pricing	Quotation (based on area)	$3000 for one unit + Installation Fees	Quotation	Quotation

Section 7: Financial Projections

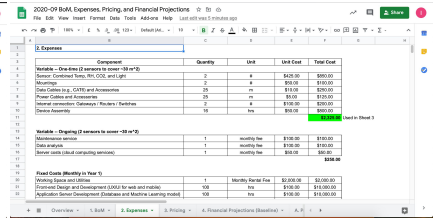


Financial Model Process

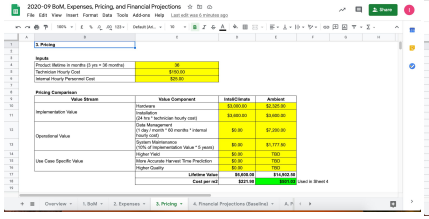
1. BoM



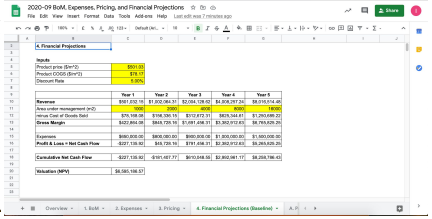
2. Expenses



3. Pricing



4. Financial Projections



Hardware Costs

Variable Expense

COGS and Price

NPV and Funding Gap

Pricing: TEV and PEV

Value Stream	Value Component*	Best Alternative	Ambient
Implementation Value	Sensing Hardware	$3,000.00	$2,325.00
	Installation (24 hours * $150/hr technician cost)	$3,600.00	$3,600.00
Operational Value	Data Management (1 day / month * 36 months product lifetime * $25/hr personnel cost)	$0.00	$7,200.00
	System Maintenance (10% of Implementation Value * 5 years)	$0.00	$1,777.50
Use Case Specific Value	Higher Yield	$0.00	TBD
	More Accurate Harvest Time Prediction	$0.00	TBD
	Higher Quality	$0.00	TBD
	Total Economic Value (TEV)	**$6,600.00**	**$14,902.50**
	TEV per m2 of farming area	**$221.90**	**$501.03**

*Main Assumptions: product lifetime and subscription contract of 3 years (36 months); 1 farming unit covers 30m2.

Financial Projections: Baseline

Inputs

Product price ($/m^2)	$500.00
Product COGS ($/m^2)	$80.00
Discount Rate	5.00%

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$500,000.00	$1,000,000.00	$2,000,000.00	$4,000,000.00	$8,000,000.00
Area under management (m2)	1000	2000	4000	8000	16000
minus Cost of Goods Sold	$80,000.00	$160,000.00	$320,000.00	$640,000.00	$1,280,000.00
Gross Margin	$420,000.00	$840,000.00	$1,680,000.00	$3,360,000.00	$6,720,000.00
Expenses	$650,000.00	$800,000.00	$900,000.00	$1,000,000.00	$1,500,000.00
Profit & Loss = Net Cash Flow	-$230,000.00	$40,000.00	$780,000.00	$2,360,000.00	$5,220,000.00

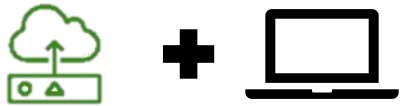
	-$230,000.00	-$190,000.00	$590,000.00	$2,950,000.00	$8,170,000.00

Valuation (NPV)	$6,522,611.32

Section 8: Milestones



Product Roadmap

and Platform Aug 2021	ML-enabled Optimization Jan 2022	Controls Aug 2022
First product milestone, which is also the MVP, will consists of a wireless network of sensors and a platform to display this data in one place.	Second product milestone will include a layer of added intelligence to the existing system in order to provide recommendations for enhancing operations.	Third product milestone will include a layer of controls that will enable fine-tuning of climate parameters remotely and automatically to create optimal conditions.
 Sensors + Visibility	 Sensors + Visibility Recommendations	 Sensors + Visibility + Actuators Recommendations Automation

Section 9: Traction



Section 10: Team

Thank you

Appendix

MVP: Two Paths

Path	Original Equipment Manufacturer (OEM)	System Integrator
Activities	1. Design IoT device based on 2. Procure components 3. Assemble device with casing 4. Develop firmware 5. Install on-site	1. Procure controller and sensors 2. Install on-site 3. Connect with existing systems
Advantages	• Flexible customization • Financially sustainable	• Ready to use • Manufacturer support
Disadvantages	• Electronics design cost and complexity • Manufacturing and distribution costs	• Solutions are wired • No data analytics capabilities • Manufacturers are far away


Device Enclosure


Temp sensor

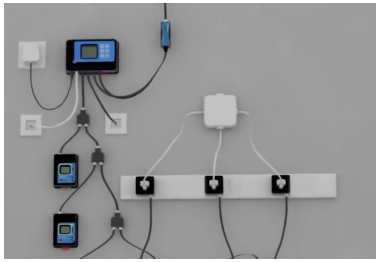

CO2 sensor


Micro-controller (“the platform”)


RH sensor



Transceiver


Connected controller and sensors


Controller


Sensors

Standards and Certifications

Singapore	Europe and US	International
IMDA: ● TR 47 : 2016 IoT reference architecture for Smart Nation ● TR 50 : 2016 IoT information and services interoperability for Smart Nation ● IMDA IoT Cyber Security Guide (2019) ● MTCS Certification Scheme	CISPR 22 standard for Information Technology Equipment-Radio Disturbance Characteristics-Limits and Methods of Measurement. More specifically: ● EN55022 ● EN61000-4-2 ● EN61000-4-3 ● EN55024 CE marking Federal Communications Commission (FCC) ● Part 15 Class A	ISO/IEC: ● 90003:2014: Software engineering ● 27000:2018: Information technology — Security techniques — Information security management systems

Draft: Patent Landscape

